

Project Charter — v0.1

Project Title

AI-Driven Project Success & Risk Decision Support System

Working product name

ProjectIQ

The name is provisional. We can change it after the prototype.

1. Project Purpose

Develop a working proof-of-concept AI/data-analytics system that uses publicly available project data to:

1. identify factors associated with project success/failure,
2. estimate project success or risk,
3. explain the major drivers behind the prediction, and
4. translate analytical results into actionable management recommendations.

The prototype will serve simultaneously as:

- PGDM Data Science dissertation project,
 - portfolio project,
 - demonstration of AI/product/project-management capability,
 - foundation for a potential AI SaaS product.
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2. Business Problem

Project managers frequently have to make decisions regarding project selection, resources, schedules and risk while working with incomplete information.

The proposed system investigates whether historical project data can be used to identify patterns associated with project outcomes and provide an explainable decision-support mechanism.

Entrepreneurial problem

A future SaaS version could help small organizations and startups answer:

"Given what we know about this project, where are the major risks and what should management do about them?"

The dissertation will validate the analytical foundation before attempting to build the complete SaaS.

3. Research Problem

Primary research question

Can publicly available project data and machine-learning techniques be used to develop an explainable decision-support framework for assessing project success and identifying project risks?

Secondary questions

1. Which project characteristics are most strongly associated with project outcomes?
 2. Which machine-learning approach provides the most useful predictive performance?
 3. Can model explainability identify actionable project-risk factors?
 4. How can predictive insights be translated into managerial recommendations?
 5. How could such a framework be developed into an AI-enabled SaaS decision-support product?
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4. Project Objectives

Academic objectives

- Conduct secondary-data research.
- Perform exploratory and statistical analysis.
- Apply appropriate machine-learning techniques.
- Compare model performance.
- Interpret findings from a management perspective.
- Develop recommendations.

Technical objectives

- Acquire and clean a suitable public dataset.
- Build a reproducible Python analysis pipeline.
- Perform feature engineering.
- Train baseline and advanced models.
- Evaluate model performance.
- Implement explainability where feasible.
- Create a basic interactive dashboard/application.

Entrepreneurial objectives

- Validate whether the underlying problem is worth solving.
- Create an MVP rather than merely a theoretical model.
- Identify potential SaaS functionality.
- Design a roadmap from prototype → product.

5. Scope

In scope

Data

- Publicly available secondary data.
- Project/product/organizational variables.
- Historical outcomes.

Analytics

- Data cleaning
- EDA
- Descriptive statistics
- Feature engineering
- Predictive modeling
- Model comparison
- Feature importance/explainability
- Business interpretation

Product

- Basic dashboard/interface
- Project risk/success assessment
- Recommendation layer
- Prototype workflow

Research

- Literature review
- Research methodology
- Hypotheses where appropriate
- Results
- Limitations
- Recommendations

Explicitly out of scope for 24 Aug

- Production SaaS
- User authentication

- Payment system
- Mobile application
- Multi-agent architecture
- Complex autonomous agents
- Large-scale cloud deployment
- Real-time enterprise integrations
- Extensive primary research

This scope protection is essential.

6. Target Users

Primary

- Project managers
- Product managers
- Startup founders
- Operations managers

Future SaaS customers

Small and medium-sized organizations that lack sophisticated project analytics infrastructure.

7. MVP

The **17 August MVP** should do only this:

User enters/selects project information



Data preprocessing



ML prediction model



Success/Risk prediction



Explainable factors



Management recommendations

Example demo

Project

- Duration: 9 months
- Team size: 6
- Complexity: High
- Budget pressure: Medium
- Resource availability: Low

Output

Predicted risk: HIGH

Major contributing factors

1. Resource availability
2. Project complexity
3. Schedule pressure

Recommended intervention

Review resource allocation and divide the project into smaller milestones.

The actual variables and outputs will depend entirely on the dataset we identify.

8. Technology Stack

Keep it familiar.

Layer	Technology
Data	Public dataset
Programming	Python
Analysis	Pandas, NumPy
Visualization	Matplotlib / Plotly
ML	Scikit-learn
Explainability	SHAP, if practical
Dashboard	Streamlit
Development	Jupyter / VS Code
Version control	Git + GitHub

Layer	Technology
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Future deployment Cloud

No need to learn a new ecosystem during these three days.

9. Three-Day Critical Path

This is the most important section.

DAY 1 — 14 AUGUST

Objective: Lock the foundation

Estimated focused work: 7–9 hours

Activity	Time
Finalize problem statement	30 min
Finalize research question/objectives	30 min
Dataset discovery	1.5 hr
Dataset evaluation	1 hr
Acquire dataset	30 min
Data dictionary	30 min
Initial data cleaning	1 hr
Initial EDA	1.5 hr
Research/literature search	1 hr
GitHub/project structure	30 min
Document decisions	30 min

End-of-day deliverable

Dataset selected + research design locked + first analytical results.

No changing the fundamental topic after Day 1 unless the dataset is unusable.

DAY 2 — 15 AUGUST

Objective: Build the analytical engine

Estimated focused work: 8–10 hours

Activity	Time
Data preprocessing	1 hr
Feature engineering	1 hr
Baseline model	1 hr
Model 2	1 hr
Model 3	1 hr
Evaluation/comparison	1 hr
Feature importance	45 min
Explainability	1 hr
Visualizations	1 hr
Begin Streamlit interface	1–1.5 hr

End-of-day deliverable

A functioning notebook:

Raw Data



Cleaning



Features



Train/Test



Models



Evaluation



Interpretation

And ideally the first version of the application.

DAY 3 — 16 AUGUST

Objective: Turn analysis into a demonstrable product

Estimated focused work: 8–10 hours

Activity	Time
Finish Streamlit MVP	2 hr
Connect model to interface	1 hr
Recommendation logic	1 hr
Improve visualizations	1 hr
Test multiple scenarios	1 hr
Prepare results	1 hr
Prepare 8–12 slide demo	1 hr
Prepare professor questions	30 min
GitHub cleanup/documentation	30 min
Demo rehearsal	1 hr

End-of-day deliverable

ProjectIQ v0.1

Something you can actually demonstrate.

10. 17 AUGUST Demo

The professor does **not** need to see a finished dissertation.

We should present:

Slide 1 — Problem

What problem are we solving?

Slide 2 — Why it matters

Business + project-management relevance.

Slide 3 — Research question

What exactly are we investigating?

Slide 4 — Dataset

Source, size, variables.

Slide 5 — Initial insights

3–5 interesting findings.

Slide 6 — Proposed architecture

Data → Analytics → ML → Explainability → Recommendations

Slide 7 — Working prototype

Live demo.

Slide 8 — Initial results

Model performance + important factors.

Slide 9 — Entrepreneurial opportunity

How this could become an AI SaaS product.

Slide 10 — Dissertation plan

How the prototype becomes the final dissertation.

Slide 11 — Limitations

Be transparent about secondary data.

Slide 12 — Feedback required

Ask professors specifically about:

- research scope
- dataset suitability
- methodology
- organizational context
- analysis depth
- recommendations

This is much better than simply asking, "Is my project okay?"

11. Risk Register

This is where we can demonstrate actual project-management thinking.

Risk	Probability	Impact	Mitigation
Dataset unsuitable	High	High	Evaluate multiple datasets Day 1
Dataset too small	Medium	High	Maintain backup dataset
ML model performs poorly	Medium	Medium	Use baseline + multiple algorithms
Missing variables	High	Medium	Clearly document limitations

Risk	Probability	Impact	Mitigation
Streamlit integration fails	Medium	Medium	Notebook remains primary MVP
Time overrun	High	High	Freeze scope after Day 1
Dissertation becomes too broad	High	High	One core research question
Professor rejects approach	Medium	High	Demo on 17th and incorporate feedback
Technical skill gap	Medium	Medium	Use established libraries/tutorial documentation
Publication quality insufficient	Medium	Low	Treat publication as bonus

12. Success Criteria

By **17 August**, success means:

Minimum

- Suitable public dataset
- Clear research question
- Clean dataset
- EDA
- At least one working ML model
- Initial results
- Prototype/demo
- Initial dissertation structure

Target

- 2–4 models compared
- Feature importance
- Explainability
- Interactive Streamlit interface
- Recommendation layer
- GitHub repository
- 10–12 slide presentation

Stretch

- Strong model performance

- SHAP
- polished dashboard
- preliminary SaaS architecture
- research-paper-quality methodology

Stretch goals must never threaten the minimum deliverables.

13. Project Management Metrics

Since you specifically want to use this to practice project management, I would maintain a small project log.

Every day record:

Metric	Example
Planned hours	8
Actual hours	7.5
Tasks planned	8
Tasks completed	7
Blockers	Dataset issue
Decisions	Dataset B selected
Risks	Model imbalance
Next action	Build baseline
Scope changes	Removed feature X

This eventually becomes **evidence in your dissertation and portfolio** that you actually managed the project.

14. Versioning

We should explicitly treat the project as a product.

v0.1 — 17 Aug

Proof of concept

v0.2 — ~20 Aug

Professor feedback incorporated

v0.3 — ~22 Aug

Final analytical prototype

v1.0 — 24 Aug

Dissertation submission version

v2.0 — After PGDM

Actual entrepreneurial MVP

This is important because **we do not need to pretend that the 24 Aug product is finished.**

The dissertation documents the first development cycle.

15. Product Roadmap After the Dissertation

Eventually:

v0.1

Prediction



v0.2

Prediction + Explainability



v0.3

Prediction + Recommendations



v1.0

AI Project Decision Support



v2.0

AI Project Copilot



v3.0

Agentic Project Management SaaS

The agent could eventually:

- monitor project data
- identify emerging risks
- generate status reports

- recommend corrective actions
- create/update tasks
- notify stakeholders
- analyze project communication
- forecast delays